

Questions

MathonGo

Q1 - 2024 (06 Apr Shift 2)

In a triangle ABC, $BC = 7$, $AC = 8$, $AB = \alpha \in \mathbb{N}$ and $\cos A = \frac{2}{3}$. If $49 \cos(3C) + 42 = \frac{m}{n}$, where $\gcd(m, n) = 1$, then $m + n$ is equal to _____

Questions

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Answer Key

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Solutions

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Q1

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\frac{2}{3} = \frac{8^2 + c^2 - 7^2}{2 \times 8 \times c}$$

$$C = 9$$

$$\cos C = \frac{7^2 + 8^2 - 9^2}{2 \times 7 \times 8} = \frac{2}{7}$$

$$49 \cos 3C + 42$$
$$49 (4 \cos^3 C - 3 \cos C) + 42$$

$$49 \left(4 \left(\frac{2}{7} \right)^3 - 3 \left(\frac{2}{7} \right) \right) + 42$$

$$= \frac{32}{7}$$

$$m + n = 32 + 7 = 39$$

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